Local Market Structure in Brno: Supermarkets and Petrol Stations

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1 Motivation

Many retail markets are local: consumers rarely drive more than a few minutes to buy groceries or fuel. Competition authorities therefore assess mergers of retail chains market by market, around each store. The European Commission's Market Definition Notice (2024, paras 72–74) describes the standard tool: a *catchment area* drawn around each supplier or customer location, either as a travel distance or as a travel time (an *isochrone*), typically covering about 80% of sales. Two recent cases show how this works in practice:

- In MOL/OMV Slovenija (M.10438, 2023), the Commission drew 15-minute driving-time isochrones around each of the parties' petrol stations, computed presence-based local market shares, and flagged an area as problematic if the parties' joint share exceeded 40% with an increment of more than 5 percentage points. It found 85 problematic areas out of 147; the divestiture of 39 stations cured all of them.
- In Carrefour/Cora–Match (25-DCC-56, 2025), the French Autorité de la concurrence used 30-minute isochrones for hypermarkets and 15-minute isochrones for supermarkets, and presumed no harm below a 25% share, or between 25% and 50% if at least three rival national banners remained.

This report applies the same tools to Brno with freely available data. It has three aims: to show how much the choice of catchment area changes measured concentration; to run a hypothetical supermarket merger through the two screens; and to check whether petrol prices are higher where local concentration is higher.

The main findings are as follows. First, the catchment definition drives measured concentration: 69.1% of the supermarkets in the Brno metropolitan area look concentrated (an HHI above 2,500) within a 1 km radius, but only 5.3% under the main definition (10-minute isochrones, 15 minutes for hypermarkets). Second, a hypothetical merger of Albert and REWE (Billa, Penny) fails the MOL/OMV screen in 60 of 69 local areas, which 14 divestitures would cure, but the French screen in only 13; the two screens react in opposite directions to the size of the catchment area. Third, in this small cross-section of petrol prices we find no statistically significant relation between local concentration and prices, although effects of up to about 0.3 CZK per litre (Natural 95) or 0.4 CZK (diesel) for a one-standard-deviation change in the number of rival groups cannot be ruled out; the owner group of a station explains about half of the variation in Natural 95 prices.

2 Data

Store locations. All supermarkets and petrol stations come from OpenStreetMap (OSM), downloaded through the Overpass API on 13 October 2026 (© OpenStreetMap contributors, ODbL). The supermarket download covers the Brno metropolitan area (Brno city and the Brno-venkov district) and its surroundings: 213 OSM objects, 163 of them in the metropolitan area, most of them with a building footprint. After removing duplicate OSM objects of the same store, convenience-format shops (such as Žabka) and the one cash-and-carry outlet (Makro), 202 supermarkets remain. The 152 in the metropolitan area are the focal outlets (Table 1 and Figure 2); the other 50 count only as competitors. The petrol-station sample covers the South Moravian Region plus a 10 km buffer, so that stations just across the regional, Slovak and Austrian borders count as competitors: 526 OSM objects, of which 384 are in the region and 156 in the Brno metropolitan area. We remove 24 duplicate objects: copies of the same station within 30 m (50 m if one of them is a building outline), and unnamed or generically named copies (an old point, a later outline, a register import) within 120 m of a station on the same road. After also removing 23 stations that sell only LPG or CNG or serve an airfield, 479 stations remain (Table 2 and Figure 1): 343 in the region, which are the focal outlets, 136 of them in the metropolitan area, and 136 in the buffer, 44 of them in Austria or Slovakia. 20 are motorway service stations, 14 of them in the region: a station counts as one if it lies in a mapped motorway services or rest area within 400 m of a motorway, or within 100 m of a motorway that is its nearest road, with no other road within 100 m.

Ownership. Brands are grouped by owner as of October 2026: Albert (Ahold Delhaize); Billa and Penny (REWE); Lidl and Kaufland (Schwarz group); Tesco; Globus; and the consumer cooperatives (Coop, Jednota, Konzum), pooled into one group. All other supermarkets are independent, and each of them counts as its own owner. For fuel: Orlen, including Benzina; MOL, including Slovnaft; Shell; OMV; EuroOil (ČEPRO, including Robin Oil); Tank ONO; and the petrol stations of hypermarkets. Many petrol stations are independent; 111 OSM stations carry neither a brand nor a name nor an operator. 80 of them are in the region, and 64 of these come from a bulk import of the ZABAGED/MPO station register into OSM in July 2026.

Road network and buildings. Roads and buildings come from the Geofabrik extract of Czechia (OSM data of 12 October 2026). Driving times use the drivable OSM road network of the region plus 12 km, with speed limits (missing limits are filled with Czech defaults by road type), processed with OSMnx (Boeing, 2017); the largest strongly connected part of the network has 93,667 nodes and 228,574 road segments. The extract ends at the national border, so Austrian and Slovak roads are missing. Four Austrian or Slovak stations lie within 250 m of a Czech road at a border crossing and are attached to it; the other 40 cannot be reached on the network, so they count only in the radius definitions and receive no predicted demand, and drive-time isochrones stop at the border. The building footprints of the same extract (317,534 in the metropolitan area plus 2 km) are used for the population grid (Section 4).

Fuel prices. Current station prices come from ceskybenzin.cz, where drivers report prices; the site has no robots.txt and no terms of use restricting access. On 13 October 2026, 81 stations in the region had a reported price (Natural 95 at 71 stations, diesel at 47, LPG at 6), each with the date of the report. Prices older than seven days are flagged. Each price station is matched to an OSM station: a station of the same brand within 300 m or, failing that, the nearest station within 150 m. One listed station (Zaris Jemnice) lies just outside the region. The analysis sample therefore has 70 stations with a Natural 95 price, 46 with a diesel price and 5 with an LPG price. For one Globus and two Tank ONO stations, the official price lists replace the crowd-sourced prices (Globus: the same prices; Tank ONO: diesel at 45.50 instead of 44.90 CZK); they add no stations. For 10 Natural 95 and 5 diesel stations, OSM and the price site

show different brands.

Table 1: Supermarkets in the Brno metropolitan area by owner group

Owner group	Brno city	Rest of metro area	Total	of which hypermarkets
Ahold Delhaize (Albert)	31	5	36	4
REWE (Billa, Penny)	24	9	33	0
Schwarz (Lidl, Kaufland)	19	6	25	5
Tesco	5	4	9	5
Globus	1	0	1	1
Coop (consumer cooperatives)	2	1	3	0
Independent and other	33	12	45	1
Total	115	37	152	16

Focal outlets: OSM supermarkets in Brno city and the Brno-venkov district, without duplicate objects, convenience formats and cash-and-carry. Hypermarkets as defined in Section 3. Independent and other: each outlet is its own owner. Source: OpenStreetMap contributors, 13 October 2026.

Table 2: Petrol stations by owner group and area

Owner group	Brno metro area	Rest of the region	Outside the region (10 km buffer)	Total
Orlen (incl. Benzina)	16	23	8	47
MOL (incl. Slovnaft)	11	14	10	35
Shell	13	14	6	33
OMV	13	13	7	33
EuroOil (ČEPRO, incl. Robin Oil)	7	14	17	38
Tank ONO	2	0	0	2
Hypermarket stations	3	1	0	4
Named independents and other brands	41	78	57	176
Unknown (no name, brand or operator)	30	50	31	111
Total	136	207	136	479

OSM petrol stations without duplicate objects, gas-only (LPG, CNG) and airfield stations. The 343 stations in the South Moravian Region (metro area and rest of the region) are the focal outlets; the 10 km buffer includes 44 stations in Austria and Slovakia. Hypermarket stations: Globus, Makro, Kaufland and Tesco, each its own owner. Named independents: each station is its own owner. Source: OpenStreetMap contributors, 13 October 2026.

3 Catchment areas

A catchment area is the set of locations that are relevant to a given outlet. We use two families of definitions. The first is a straight-line radius around the outlet:

$$A_i^{\text{radius}}(r) = \{\, x : d(x, \ell_i) \le r \,\}, \qquad r \in \{1, 2, 3\} \text{ km},$$

$x:$ a location, $\quad \ell_i:$ location of outlet i, $\quad d:$ straight-line distance.

The second is a driving-time isochrone:

$$A_i^{\text{drive}}(\tau) = \{\, x : t(x, \ell_i) \le \tau \,\},$$

$t(x, \ell_i):$ shortest driving time on the road network from location x to outlet i.

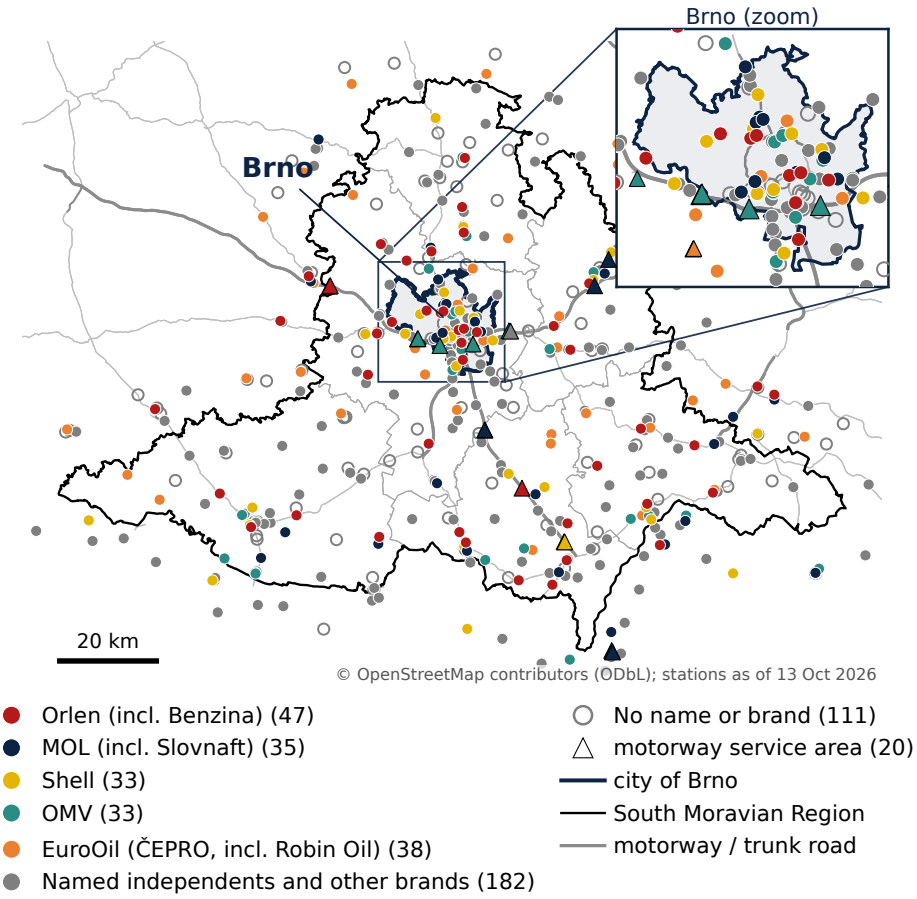


Figure 1: Petrol stations in the South Moravian Region (plus 10 km), by brand group

All 479 stations. Filled grey circles: named independents and other brands, including Tank ONO and the hypermarket stations; open circles: stations without a name, brand or operator in OSM; triangles: motorway service stations (in a motorway services area or reached from the motorway). The shaded area is the city of Brno, enlarged in the inset. Source: OpenStreetMap contributors, 13 October 2026.

For petrol stations we use 5 and 10 minutes, and 15 minutes as in MOL/OMV; for supermarkets 10 minutes, and 15 minutes for hypermarkets, following the French practice (the main definition), and as robustness 5, 10 and 15 minutes for all stores. A store counts as a hypermarket if it carries a hypermarket banner (Globus, Kaufland), has its own building of at least 2,500 m² (except Billa, Lidl and Penny, which run only supermarket and discount formats in Czechia), or is an Albert or Tesco anchoring a shopping centre of at least 10,000 m². This gives 16 hypermarkets in the metropolitan area. A plain rule of a 2,500 m² footprint or a hypermarket banner would give 40, mostly because large Billa, Lidl and Penny buildings or whole shopping-centre buildings would count.

Outlets and customer locations are attached to the nearest point of the nearest road segment, not to the nearest intersection, because many petrol stations lie on long road segments between intersections. In the road network, the isochrone is the set of road segments reachable within the time limit, including the reachable part of the segments at the limit, buffered by 150 m and merged into one polygon. To keep this fast for 1,923 isochrones, the buffers are merged on a 25 m raster and traced back to polygons, which puts the boundary within about 15 m of the exact one. The median 10-minute isochrone of a supermarket covers 97.6 km² (5 minutes: 21.7 km²; 15 minutes: 213.3 km²), and the median 5-minute isochrone of a petrol station covers 16.5 km² (10 minutes: 65.1 km²); for comparison, radii of 1, 2 and 3 km cover 3.1, 12.6 and 28.3 km².

The outlets that compete with an outlet are those located inside its catchment area:

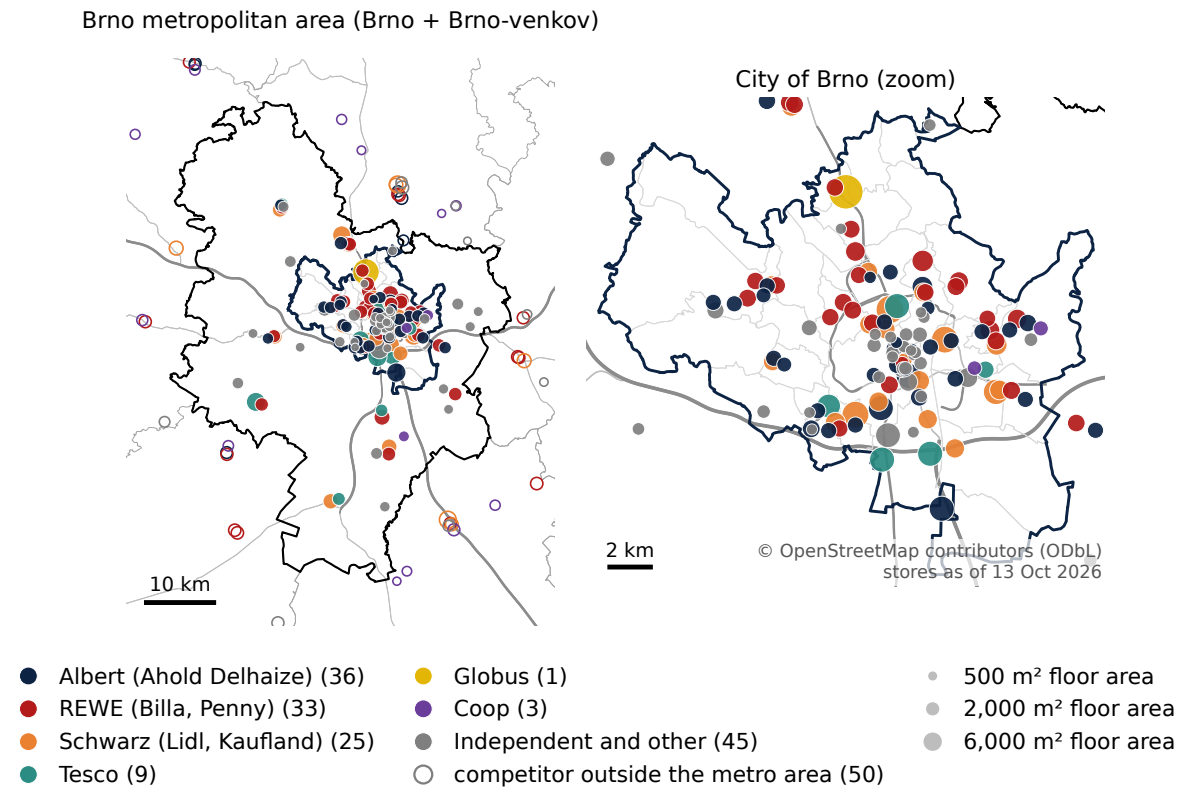


Figure 2: Supermarkets in the Brno metropolitan area, by owner group
The 152 focal supermarkets (filled) and the 50 competitors outside the metropolitan area (open circles). Marker size is proportional to the floor-area proxy of Section 4. Source: OpenStreetMap contributors, 13 October 2026.

$$N_i = \{\, j : \ell_j \in A_i \,\}, \qquad N_i : \text{ competitor set of outlet } i.$$

For the drive-time definitions this means that one outlet competes with another if the driving time from the first to the second is within the limit; we compute the competitor sets from these network times directly. The isochrone polygons on the maps are only an approximate picture of these sets, because the 150 m buffer and the reachable parts of the segments at the limit take in outlets just beyond it. For 10-minute supermarket catchments the two competitor sets overlap by 95.0% (Jaccard index: rivals in both sets over rivals in either set); 99.1% of the rivals found from network times lie inside the polygon, and 95.9% of the rivals inside the polygon are found from network times. For 5-minute petrol-station catchments the overlap is 80.3% (96.0% and 83.0%): the polygons take in 425 rival stations that are not in the network-time sets, a median of 5.6 minutes away.

Supplier-centric versus customer-centric catchment areas. Catchment areas can be drawn around suppliers or around customers (Market Definition Notice, para. 73). The definition above is *supplier-centric*: the area is drawn around each store, and its competitors are the stores inside it. Two outlets are direct competitors if one lies in the other's catchment area; outlets further apart can still constrain each other through a chain of overlapping areas (para. 87). Figure 3 shows three overlapping areas south of Brno. The *customer-centric* approach draws the area around each customer location instead, and asks which stores a customer there can reach. With the population grid, the customer locations are the grid cells:

$$M_c = \{\, j : t(c, \ell_j) \le \tau \,\}, \qquad \mathrm{HHI}_c = 10\,000 \sum_f s_{fc}^2, \qquad \overline{\mathrm{HHI}} = \frac{\sum_c \mathrm{Pop}_c \, \mathrm{HHI}_c}{\sum_c \mathrm{Pop}_c},$$

$c:$ grid cell, $\quad \mathrm{Pop}_c:$ its residents, $\quad s_{fc}:$ share of owner group f in M_c,

where the shares are computed among the outlets that a customer in the cell can reach, with the same weights as before. Instead of counting problematic stores, the customer-centric merger screen counts the residents who would face a joint share above 40% with an increment above 5 percentage points among the stores they can reach. Since a cell belongs to neither party, the increment is the smaller of the two parties' shares:

$$S_c = s_{ac} + s_{bc}, \qquad \Delta_c = \min(s_{ac}, s_{bc}), \qquad a, b: \text{ the merging owner groups.}$$

Which approach fits depends on how firms set prices. If suppliers do not price by customer location, the Commission usually draws catchment areas around suppliers; if they negotiate customer-specific prices or deliver to customers, it prefers areas around customers (para. 73), as for industrial chocolate (Cargill/ADM Chocolate Business, 2015) and beverage cans (Ball/Rexam, 2016). Supermarkets and petrol stations post the same price to everyone who comes to the store, and authorities accordingly use store-centred areas: the Commission in MOL/OMV, the French authority in Carrefour/Cora, the UK Competition and Markets Authority in Sainsbury's/Asda (2019; where survey data on where customers come from shaped and weighted the areas, but the areas were still centred on the parties' stores). Customer-centric analyses of grocery markets appear mainly in research (Ellickson, Grieco and Khvastunov, 2020) and in parties' submissions; in MOL/OMV the Commission did not reject a customer-centric approach in principle, but found the parties' version unreliable. We therefore use the supplier-centric approach throughout and report the customer-centric measures for supermarkets as a robustness check.

4 Measuring local concentration

Local market shares. The share of owner group f in the catchment area of outlet i is

$$s_{fi} = \frac{\sum_{j \in N_i} w_j \, \mathbb{1}[g(j) = f]}{\sum_{j \in N_i} w_j},$$

$g(j):$ owner group of outlet j, $\quad w_j:$ weight of outlet j,

where the weight stands in for the outlet's sales, which we do not observe.

Weights without sales data. We use three weighting schemes:

1. *Equal weights* (baseline): every outlet counts once. This is the presence-based approach in its simplest form.
2. *Floor area* (supermarkets only): the weight is the footprint of the store's own building in OSM, a proxy for sales area. The French authority computes shares in sales area. For stores inside a larger building such as a shopping centre, without a footprint, or with an implausibly small one (61 of the 202 stores), the weight is the median own-building footprint of the same banner, capped at the footprint of the enclosing building.
3. *Predicted demand* (both sectors): residents of each location k spread their demand over the outlets within reach, with closer outlets receiving more (a Huff or gravity model). With the population of each location and the driving times from it to the outlets,

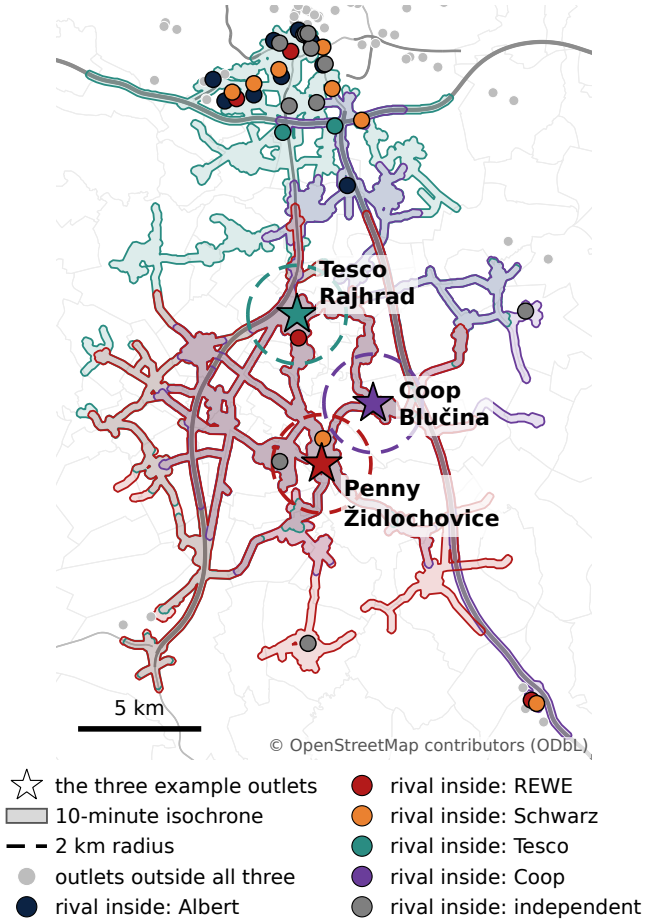


Figure 3: Example of overlapping catchment areas: three supermarkets south of Brno

Stars: Tesco (Rajhrad), Coop (Blučina) and Penny (Židlochovice), each with its 10-minute driving-time isochrone (shaded) and a 2 km radius (dashed). Coloured dots: rival supermarkets inside at least one of the three isochrones, by owner group; grey dots: supermarkets outside all three. The isochrones follow the roads and reach much further along the main roads than the radius. Source: OpenStreetMap contributors (stores 13 October 2026, roads 12 October 2026).

$$\pi_{kj} = \frac{\exp(-\lambda\, t_{kj})}{\sum_{l \in J_k} \exp(-\lambda\, t_{kl})}, \qquad w_j = \sum_k \mathrm{Pop}_k\, \pi_{kj},$$

k : location, $\quad t_{kj}$: driving time from k to outlet j, $\quad J_k$: choice set of k,

where the choice set contains the outlets within 20 minutes of the location. The distance-decay parameter is set so that about 80% of predicted demand comes from within the catchment area, in line with the Commission's 80% rule. This mimics the Commission's approach in MOL/OMV, which weighted stations by their brand's average volume; we have no volumes, but we do have where people live. We calibrate the parameter on the focal outlets, with a 10-minute catchment for all supermarkets and a 5-minute catchment for petrol stations:

$$\hat{\lambda}_{\text{supermarkets}} = 0.327, \qquad \hat{\lambda}_{\text{petrol stations}} = 0.925 \qquad \text{per minute},$$

so predicted demand halves with every 2.1 extra minutes of driving for supermarkets and every 0.75 minutes for petrol stations. The weights have no size term: a hypermarket and a small supermarket at the same place receive the same predicted demand, and across the focal supermarkets the correlation between predicted demand and floor area is only 0.03.

Where people live. Inside the Brno metropolitan area, the locations are the cells of a 1×1 km population grid built as in Martin (2024): the 2021 census population of each municipality (in Brno, of each of the 29 city districts; Czech Statistical Office) is distributed over its OSM buildings in proportion to their ground area, and summed within each cell,

$$\mathrm{Pop}_c = \sum_m \mathrm{Pop}_m \sum_{b \in B_m,\, b \subset c} \frac{\mathrm{area}_b}{\sum_{b' \in B_m} \mathrm{area}_{b'}},$$

m : municipality or city district, B_m : its buildings, area_b : ground area of building b,

where each building is counted in the cell that contains a point inside its footprint; the representative consumer of a cell sits at its centroid. The buildings are the 207,325 footprints in the metropolitan area from the Geofabrik extract, after dropping 88 footprints that lie inside or duplicate another one, so that every piece of ground counts once. The grid has 1,377 populated cells with 625,013 residents (Figure 4). Every building counts in proportion to its footprint, whatever its use and height. Czech OSM tags building uses almost everywhere, and this baseline places 39.3% of residents in buildings with a clearly non-residential tag (civic amenities, warehouses, factories, barns, garages), so it puts a good part of demand where people work or shop. As robustness we therefore use two variants: residential buildings only, and residential buildings weighted by floor area (footprint times the number of floors, which is tagged for about 80% of residential ground area); each variant has its own decay parameter. Outside the metropolitan area, the locations are municipality centroids with OSM populations: for supermarkets the 173 municipalities within 10 km of the metropolitan area, for petrol stations all 485 other municipalities of the region. There are no locations outside the region, so the predicted demand of stations in the buffer is understated; and 83 locations with 16,814 residents, mostly municipality centroids in fields or forests, lie more than 1 km from a road. The grid is produced by code/brno/03b_population_grid.py and saved as results/brno/data/clean/pop_grid_brno_1km.gpkg.

Concentration measures. For each outlet and each catchment definition we compute the Herfindahl–Hirschman index and the number of rival owner groups:

$$\mathrm{HHI}_i = 10\,000 \sum_f s_{fi}^2, \qquad R_i = \big|\{\, g(j) : j \in N_i \,\} \setminus \{g(i)\}\big|, \qquad R_i : \text{ rival owner groups},$$

as well as the driving time to the nearest rival outlet, that is, from the outlet to the nearest outlet of another owner. Each independent store or station counts as a separate owner group.

5 Hypothetical supermarket merger

We simulate a merger between two owner groups with a large presence in Brno that are not already linked, Ahold Delhaize (Albert) and REWE (Billa, Penny). Around each outlet of either party we compute the parties' joint share and the increment, that is, the share of the other party:

$$S_i = s_{ai} + s_{bi}, \qquad \Delta_i = \begin{cases} s_{bi} & \text{if outlet } i \text{ belongs to } a, \\ s_{ai} & \text{if outlet } i \text{ belongs to } b, \end{cases} \qquad a, b : \text{ the merging owner groups.}$$

An area is problematic

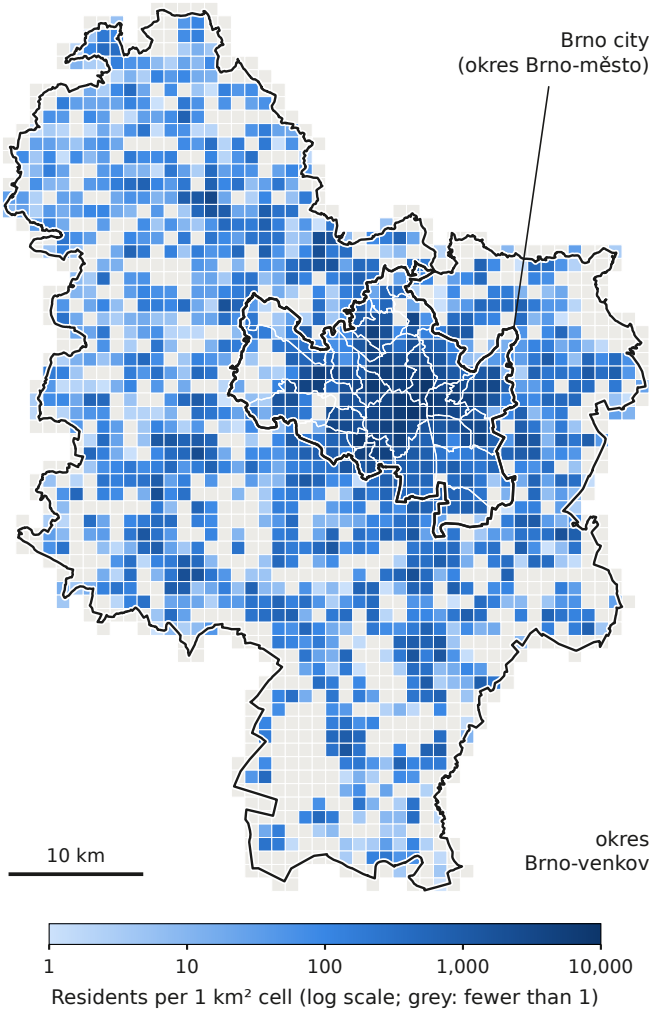


Figure 4: Building-based population grid of the Brno metropolitan area

Residents per 1 km² cell (log scale; grey: fewer than one). 2021 census populations of the municipalities of Brno-venkov and the districts of Brno, allocated to OSM buildings in proportion to their ground area. Source: Czech Statistical Office, Census 2021; OpenStreetMap contributors (Geofabrik extract, 12 October 2026).

- under the MOL/OMV screen, if the joint share is above 40% and the increment above 5 percentage points;
- under the French screen, if the joint share is above 50%, or between 25% and 50% with fewer than three rival owner groups.

In both screens the other party must be present in the area. Because every independent store counts as a rival owner group, we also report a variant of the French screen that counts only the national chains (Schwarz, Tesco, Globus, the cooperatives and the buyer of divested stores), which is closer to the French authority's "rival national banners".

Remedy. We then look for a small set of divestitures that cures every problematic area. The algorithm is greedy: it divests the party outlet that cures the most remaining problematic areas, recomputes all shares with that outlet owned by a new buyer independent of the parties, and repeats until no problematic area is left. All divested outlets go to the same buyer, and a divested outlet's own area no longer counts. The candidates are all 88 Albert and REWE supermarkets

in the data, including those just outside the metropolitan area; ties are broken by the parties' summed joint share. As in MOL/OMV, one divested outlet can cure several overlapping areas.

6 Prices and local concentration

For each petrol station with a reported price we estimate

$$p_i = \alpha + \beta\, C_i + \gamma' X_i + \mu_{g(i)} + \varepsilon_i,$$

separately for Natural 95 and diesel, where the price is in CZK per litre, the concentration measure is the number of rival owner groups within 5 minutes' drive (main specification), the HHI within 5 minutes (in thousands), the log of one plus the number of rivals within 10 minutes, the driving time to the nearest rival, or the number of rival outlets within 2 km. The controls include a motorway-service dummy, a dummy for the city of Brno and a stale-price dummy, each dropped where it does not vary in the sample; the owner-group fixed effects compare stations of the same owner group in Table 2, where named independents form one group and the hypermarket stations another. Standard errors are heteroskedasticity-robust (HC1). Search theory predicts that prices fall with the number of stations consumers compare, so we expect a negative coefficient on rival counts and a positive one on the HHI. As robustness checks we drop the stations whose brand differs between OSM and the price site as well as price outliers, ignore the unnamed register stations as rivals, count rival groups within 10 minutes, weight the HHI by predicted demand, and cluster standard errors by municipality.

This is a cross-sectional correlation, not a causal effect: stations open where demand is high, and crowd-sourced prices are noisy. Fischer, Martin and Schmidt-Dengler (2026) use entry events to estimate causal effects in the German petrol market.

7 Results

7.1 Local concentration across catchment definitions

Supermarkets. Figure 5(a) and Table 3 show how strongly measured concentration depends on the catchment definition; Appendix Table 9 gives the full distributions. A 1 km radius around one of the 152 supermarkets in the metropolitan area contains 6.4 outlets on average, including the focal one, while the main definition (10 minutes, hypermarkets 15 minutes) contains 75.2. The median number of rival owner groups rises from 2 within 1 km to 26 under the main definition, and the median HHI with equal weights falls from 3,750 to 1,539. Within 1 km, 69.1% of the supermarkets have an HHI above 2,500 and 63.2% face fewer than three rival owner groups; under the main definition, only 5.3% and 2.0% do. A five-minute isochrone gives about the same picture as a 3 km radius, which covers a similar area: a median HHI of 2,000 in both cases, and 30.9% against 34.9% of the outlets above 2,500. Short radii thus give much higher concentration than drive-time catchments, above all in Brno city, where a 10-minute drive reaches dozens of stores; if customers drive 10 minutes, as the French practice assumes, radii of 1 or 2 km overstate concentration. The nearest store of another owner is 1.15 minutes away for the median supermarket (90th percentile: 3.16 minutes).

The weights matter as well. Floor-area weights raise measured concentration, because hypermarkets dominate floor space: under the main definition the median HHI is 1,993 with floor areas against 1,539 with equal weights, and 12.5% instead of 5.3% of the supermarkets are above 2,500. Predicted-demand weights hardly change the picture (median 1,481; 5.3% above 2,500).

Petrol stations. Catchments of petrol stations contain fewer outlets (Figure 5(b), Table 3 and Appendix Table 10): 2.0 within 1 km and 7.3 within 5 minutes. Under the five-minute

definition the median HHI is 2,000; 34.7% of the 343 stations in the region have an HHI above 2,500, 33.2% face fewer than three rival owner groups, and 10.2% have no rival at all. Within 10 minutes these shares fall to 5.5%, 5.0% and 0.3%. Predicted-demand weights raise concentration markedly (median HHI 2,643 within 5 minutes; 54.2% above 2,500): with the steep calibrated decay, most of a town's demand goes to the station closest to it, so the weights differ much more between petrol stations than between supermarkets. Counting every unnamed station as a separate rival also matters. If the unnamed stations from the register import are ignored as rivals, the median number of rival groups within 5 minutes falls from 4 to 3, and the share of stations with fewer than three rival groups rises from 33.2% to 43.1%. The nearest rival is 1.68 minutes away for the median station (90th percentile: 5.53 minutes). On the choice of local market definition more generally, see Pennerstorfer and Yontcheva (2021) and Genakos and Kampouris (2026).

Table 3: Share of concentrated outlets by catchment definition (%)

Panel A: Supermarkets (152 outlets in the Brno metropolitan area)

Catchment definition	HHI > 2,500, equal weights	HHI > 2,500, floor area	HHI > 2,500, predicted demand	Fewer than 3 rival groups
1 km radius	69.1	82.9	73.0	63.2
2 km radius	47.4	76.3	48.7	37.5
3 km radius	34.9	62.5	35.5	21.1
5 min drive	30.9	58.6	34.2	15.8
10 min drive	5.9	13.8	5.9	2.6
15 min drive	0.0	4.6	0.0	0.0
10 min, hypermarkets 15 min (main)	5.3	12.5	5.3	2.0

Panel B: Petrol stations (343 outlets in the South Moravian Region)

Catchment definition	HHI > 2,500, equal weights	HHI > 2,500, predicted demand	Fewer than 3 rival groups	No rival outlet
1 km radius	89.5	96.5	87.5	51.6
2 km radius	65.6	80.8	65.0	28.6
3 km radius	49.3	70.0	47.5	15.7
5 min drive	34.7	54.2	33.2	10.2
10 min drive	5.5	14.0	5.0	0.3
15 min drive	0.9	0.9	0.6	0.0

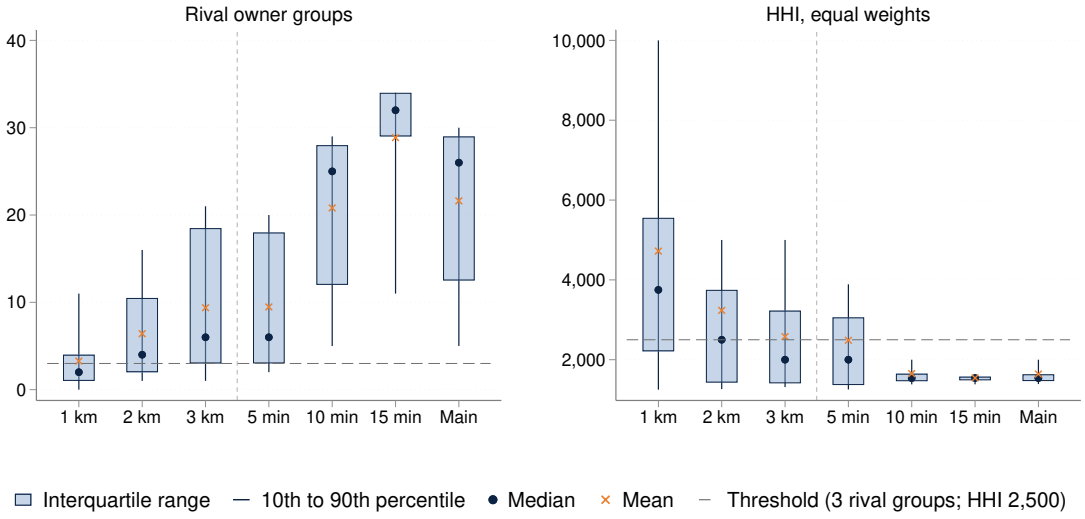
HHI computed on owner-group shares in the catchment area; predicted demand: Huff weights (Section 4). Rival groups: owner groups other than the outlet's own, every independent outlet counted separately. Source: OpenStreetMap contributors (stores 13 October 2026, roads 12 October 2026); Czech Statistical Office, Census 2021; own calculations.

7.2 Hypothetical merger of Albert and REWE

Albert and REWE together operate 69 of the 152 supermarkets in the metropolitan area (36 Albert, 33 Billa or Penny). Under the main definition, 68 of the 69 catchment areas around these stores contain a store of the other party. Table 4 and Figure 6 show the results of the two screens.

With equal weights, the MOL/OMV screen flags 60 of the 69 areas as problematic. The greedy remedy cures all of them with 14 divestitures, 5 Albert and 9 REWE stores, all of them in the metropolitan area and twelve in Brno city; as in MOL/OMV, one divestiture cures several overlapping areas. Floor-area weights flag fewer areas (44, cured by 7 divestitures), because

(a) Supermarkets: 152 outlets in the Brno metropolitan area



(b) Petrol stations: 343 outlets in the South Moravian Region

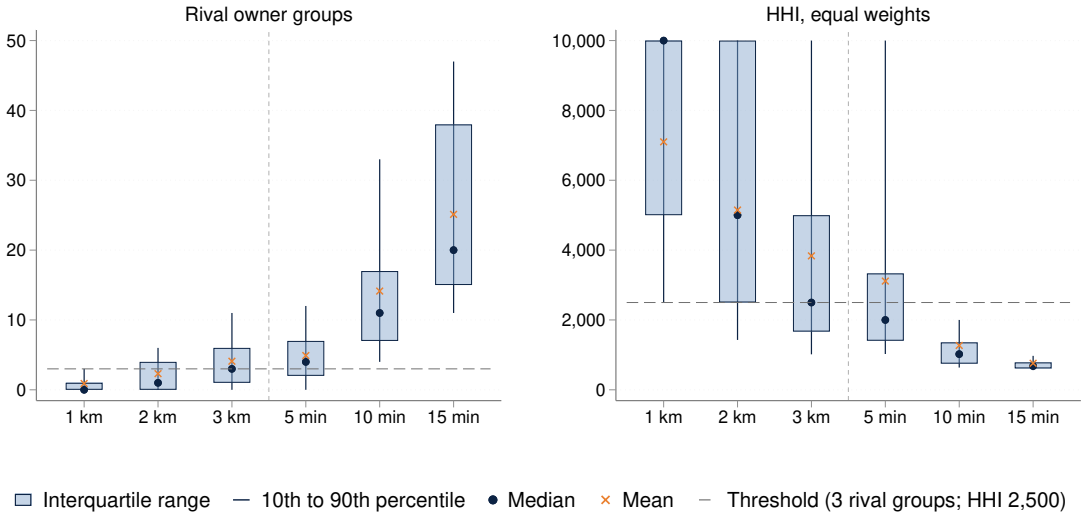


Figure 5: Rival owner groups and HHI across catchment definitions

Boxes: interquartile range; whiskers: 10th to 90th percentile; dots: median; crosses: mean. Dashed lines: three rival groups and an HHI of 2,500. Radii in km, isochrones in minutes' drive; Main: 10 minutes, hypermarkets 15 minutes. HHI with equal weights; every independent outlet is a separate owner group. Source: OpenStreetMap contributors (stores 13 October 2026, roads 12 October 2026); own calculations.

REWE runs no hypermarket and the rivals' hypermarkets dominate floor space. Predicted-demand weights give almost the same result as equal weights: 59 problematic areas and 11 divestitures, one of them (Penny in Moravský Krumlov) outside the metropolitan area. With predicted demand based on residential buildings only, or on residential floor area, the MOL/OMV screen flags 61 areas (12 divestitures).

The French screen is far more lenient: it flags 13 areas with equal weights (4 divestitures), 4 with floor areas (4) and 8 with predicted demand (5). Under the main definition every one of the 69 areas contains at least three rival owner groups, so only a joint share above 50% triggers the screen. Part of this leniency comes from counting every independent store as a rival owner group. If only the national chains count as rivals, the French screen flags 24, 18 and 20 areas.

The two screens react in opposite directions to the size of the catchment area (Table 5). Larger areas more often contain stores of both parties, and the joint share approaches the parties' share of all stores (69 of 152, or 45%), which is above the 40% threshold of the MOL/OMV screen. That screen therefore flags 30 areas with a 1 km radius but 66 with 15-minute isochrones (equal weights). Larger areas also contain more rival groups, so the French screen flags 27 areas with a 1 km radius but only 2 with 15-minute isochrones. The choice of catchment area matters as much as the choice of screen.

Table 4: Albert + REWE merger: problematic areas and divestitures, main definition

Weighting	MOL/OMV screen: problematic areas	MOL/OMV screen: divestitures	French screen: problematic areas	French screen: divestitures
Equal weights	60	14	13	4
Floor area	44	7	4	4
Predicted demand	59	11	8	5

The 69 catchment areas (10 minutes, hypermarkets 15 minutes) around the parties' supermarkets in the Brno metropolitan area. MOL/OMV screen: joint share above 40% and increment above 5 percentage points. French screen: joint share above 50%, or 25–50% with fewer than three rival owner groups (independents counted separately). Divestitures: size of the greedy divestiture set that cures all problematic areas under that screen. Source: OpenStreetMap contributors; own calculations.

Table 5: Albert + REWE merger: problematic areas by catchment definition

Catchment definition	MOL/ OMV: equal weights	MOL/ OMV: floor area	MOL/ OMV: predicted demand	French: equal weights	French: floor area	French: predicted demand
1 km radius	30	32	30	27	32	27
2 km radius	40	40	39	33	32	33
3 km radius	46	34	44	29	20	29
5 min drive	46	28	44	25	14	23
10 min drive	59	44	58	13	4	8
15 min drive	66	63	66	2	0	0
10 min, hypermarkets 15 min (main)	60	44	59	13	4	8

Number of the 69 party areas flagged as problematic under each screen and weighting scheme. Screens as in Table 4. Source: OpenStreetMap contributors; own calculations.

Customer-centric robustness. Table 6 and Figure 7 report the customer-centric measures for the 1,377 populated grid cells, which hold 625,013 residents. With a 10-minute catchment for all stores and equal weights, the population-weighted mean HHI is 1,971 (median 1,590).

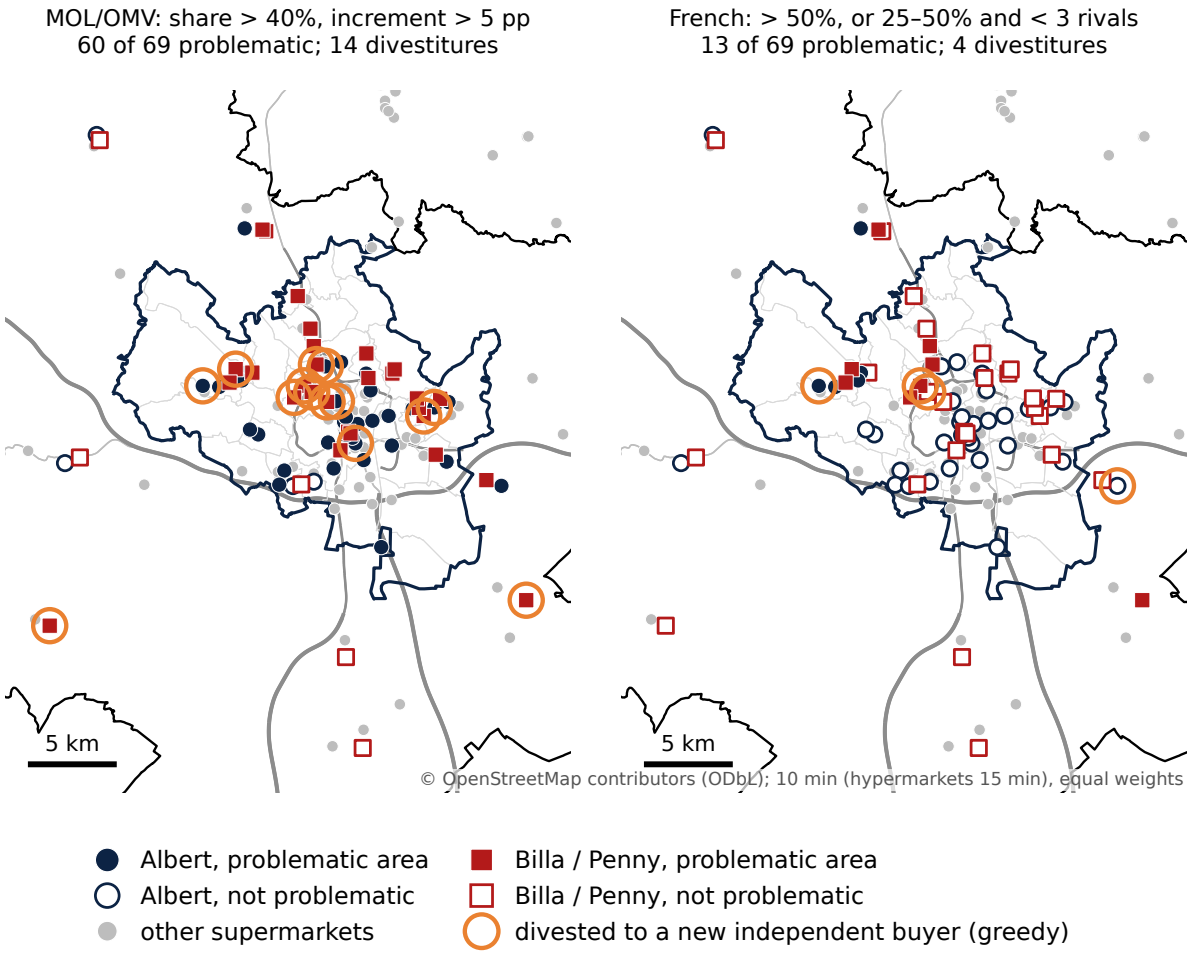


Figure 6: Albert + REWE merger: problematic areas and divestitures (main definition, equal weights) Left: MOL/OMV screen; right: French screen. Filled markers: party stores whose catchment area is problematic; open markers: not problematic; orange rings: stores divested by the greedy remedy. Source: OpenStreetMap contributors (stores 13 October 2026, roads 12 October 2026); own calculations.

This is higher than the mean of 1,652 across supermarkets for the same definition: stores cluster where competition is dense, whereas many residents in the outer parts of Brno-venkov reach only a few stores. 10,429 residents have no supermarket within 10 minutes, and 31,954 reach fewer than three owner groups. The merger would raise the parties' joint share above 40%, with an increment above 5 percentage points, for 458,495 residents (73.4%). After the 14 supplier-centric MOL/OMV divestitures, 92,857 residents (14.9%) would still be affected. Part of this gap reflects the different definition, because the divestitures were chosen with hypermarkets counted within 15 minutes: with hypermarkets counted within 15 minutes, as in the supplier-centric remedy, 31,051 residents (5.0%) remain affected. Even like for like, a remedy designed around the parties' stores does not cure every customer location. The customer-centric screen is sensitive to the definition and the weights. If hypermarkets count within 15 minutes, 421,494 residents (67.4%) are affected with equal weights but only 47,110 (7.5%) with floor-area weights, because the rivals' large hypermarkets then enter most residents' catchments. With each cell's own predicted choice probabilities as weights, the mean HHI is 2,228 and 426,376 residents are affected.

Table 6: Customer-centric measures for supermarkets

	Equal weights	Floor area	Predicted demand
Supermarkets and hypermarkets within 10 min			
Population-weighted mean HHI	1,971	2,555	1,952
Population-weighted median HHI	1,590	2,125	1,504
Residents without a supermarket in reach	10,429	10,429	10,429
Residents reaching fewer than 3 owner groups	31,954	31,954	31,954
Residents failing the MOL/OMV screen	458,495	313,149	459,534
share of all residents (%)	73.4	50.1	73.5
after the MOL/OMV divestitures	92,857	68,876	96,643
share of all residents (%)	14.9	11.0	15.5
Supermarkets within 10 min, hypermarkets within 15 min			
Population-weighted mean HHI	1,875	2,459	1,857
Population-weighted median HHI	1,569	1,925	1,509
Residents without a supermarket in reach	8,629	8,629	8,629
Residents reaching fewer than 3 owner groups	24,874	24,874	24,874
Residents failing the MOL/OMV screen	421,494	47,110	414,547
share of all residents (%)	67.4	7.5	66.3
after the MOL/OMV divestitures	31,051	5,316	39,900
share of all residents (%)	5.0	0.9	6.4

The 1,377 populated 1 km grid cells of the Brno metropolitan area (625,013 residents). Each cell's catchment area contains the supermarkets within the stated driving time; HHI averages exclude cells without a supermarket. MOL/OMV screen: the parties' joint share is above 40% and the smaller party's share above 5 percentage points among the stores in reach. Divestitures: the supplier-centric MOL/OMV remedy of Table 4 for the same weights. Source: OpenStreetMap contributors; Czech Statistical Office, Census 2021; own calculations.

7.3 Prices and local concentration

Natural 95 costs 44.94 CZK per litre on average across the 70 stations with a price (standard deviation 0.86, range 43.50–46.90), and diesel 46.69 CZK across 46 stations (standard deviation 1.47). Figure 8 maps the Natural 95 prices, and Appendix Table 11 summarises both samples. Figure 9 plots prices against the two main concentration measures after removing the controls and the owner-group fixed effects, and Table 7 reports the regressions.

We find no statistically significant relation between concentration and prices. In the main specification, one more rival owner group within 5 minutes changes the Natural 95 price by −0.004 CZK (standard error 0.033; 95% confidence interval −0.07 to +0.06 CZK) and the diesel price by −0.002 CZK (0.045). The estimates are imprecise, however: for one standard deviation more rival groups (4.56 in the Natural 95 sample, 4.31 in the diesel sample), the 95% confidence intervals run from −0.32 to +0.29 CZK per litre for Natural 95 and from −0.40 to +0.39 CZK for diesel, so with 70 and 46 stations effects of up to about 0.3 to 0.4 CZK per litre cannot be ruled out. The HHI, the number of rivals within 10 minutes, the time to the nearest rival and the number of rival outlets within 2 km are not significant either, and neither are the robustness checks in Table 8; the smallest p-value is 0.133 (diesel, rival groups within 10 minutes). No station with a price is a motorway service station, so the motorway dummy drops out.

What does explain Natural 95 prices is ownership. The owner-group fixed effects alone explain 48.3% of the variation in Natural 95 prices (adjusted R^2 0.424), and 53.6% (adjusted 0.474) without the 10 prices whose OSM station carries another brand than the price site, which may belong to another owner group. Among these 60 stations, named independents charge 44.49 CZK for Natural 95 on average, against 45.41 CZK (Orlen) to 46.53 CZK (Shell, three stations) at the majors; EuroOil (44.80), Tank ONO (43.90) and the hypermarket stations (44.50) are also

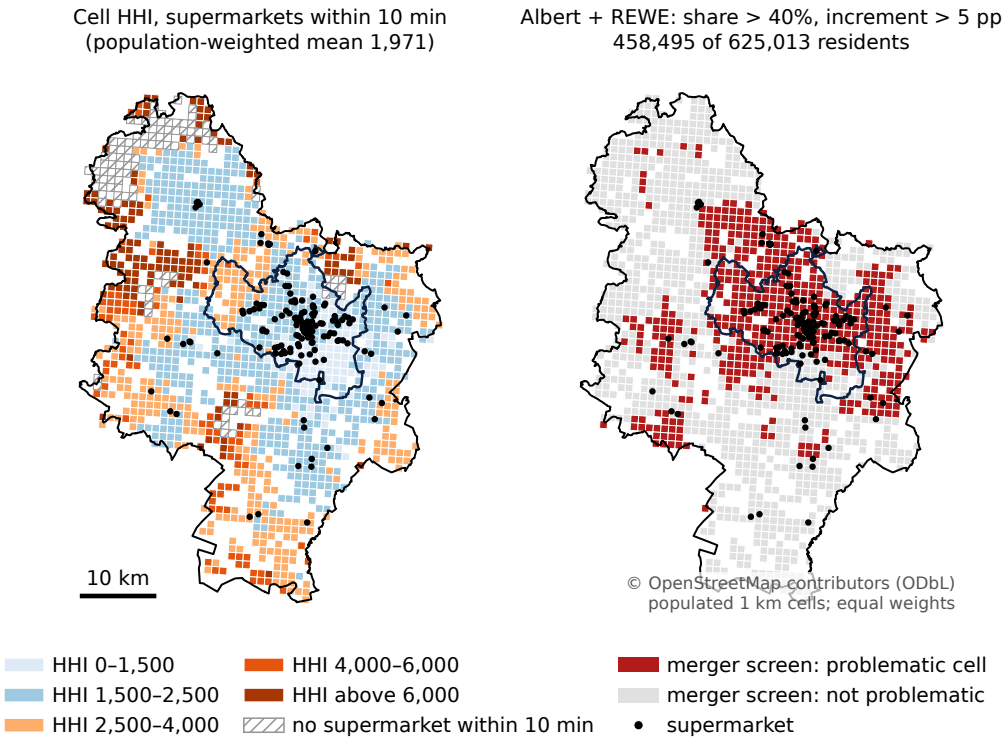


Figure 7: Customer-centric view: cell HHI and the merger screen (10 minutes, equal weights)
Left: HHI among the supermarkets within 10 minutes of each populated grid cell; hatched: no supermarket within 10 minutes. Right: cells where the merger would raise the parties' joint share above 40% with an increment above 5 percentage points. Black dots: supermarkets. Source: OpenStreetMap contributors (stores 13 October 2026, roads and buildings 12 October 2026); Czech Statistical Office, Census 2021; own calculations.

cheap. For diesel the pattern is fragile. The fixed effects explain 46.3% of the variation across all 46 diesel prices, but this rests on three MOL stations in Brno whose diesel prices (49.90 to 51.90 CZK, all reported on 11 October) are flagged as outliers and lie 4.0 to 7.4 CZK above their Natural 95 prices, against a median gap of 2.0 CZK; they may be reports of premium diesel. Without them, owner groups explain 20.3% of the variation (adjusted 7.1%) and are not jointly significant ($p = 0.20$).

These results call for caution. The samples are small (70 and 46 stations), and the regression is a cross-section: stations enter where demand and margins are high, which can mask a negative effect of competition on prices (Fischer, Martin and Schmidt-Dengler, 2026). The prices are crowd-sourced, 47 of the 70 Natural 95 prices were reported on 11 October, and 10 Natural 95 and 5 diesel stations show a different brand on the price site than in OSM. LPG prices are available for only 5 stations, too few for a regression.

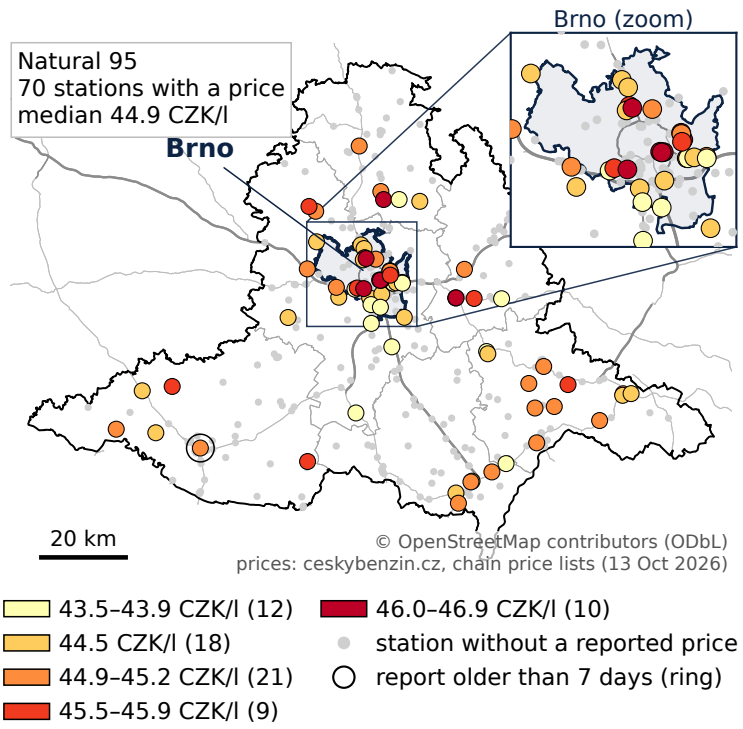


Figure 8: Natural 95 prices in the South Moravian Region

The 70 stations with a price collected on 13 October 2026 (reported 4–13 October, 47 of them on 11 October); ringed: report older than seven days; grey dots: stations without a price. No station with a price is a motorway service station. Source: ceskybenzin.cz (user reports) and the price lists of Tank ONO and Globus; OpenStreetMap contributors.

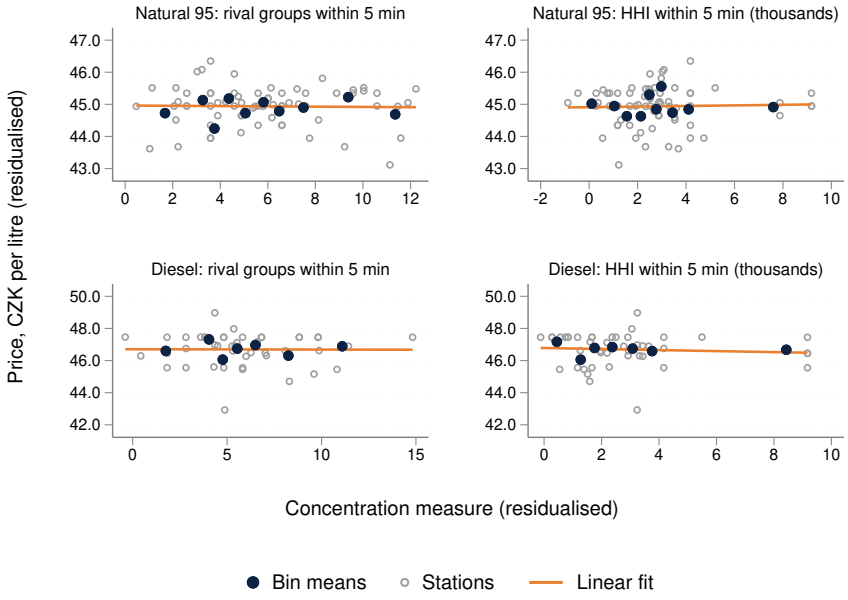


Figure 9: Station prices and local concentration (binned scatter)

Prices and concentration measures residualised on the station controls and the owner-group fixed effects, with the sample means added back; the slope of the linear fit equals the coefficient in Table 7, columns (1) and (2). Bins: ten for Natural 95 (70 stations), seven for diesel (46 stations). Source: ceskybenzin.cz (user reports), price lists of Tank ONO and Globus; OpenStreetMap contributors; own calculations.

Table 7: Fuel prices and local concentration

	(1)	(2)	(3)	(4)	(5)
Panel A: Natural 95					
Rival groups within 5 min	−0.004				
	(0.033)				
HHI within 5 min (thousands)		0.009			
		(0.026)			
Log(1 + rival outlets within 10 min)			−0.163		
			(0.153)		
Driving time to nearest rival (min)				−0.025	
				(0.043)	
Rival outlets within 2 km					0.046
					(0.045)
City of Brno	0.127	0.118	0.341	0.079	−0.072
	(0.331)	(0.260)	(0.351)	(0.262)	(0.306)
Owner-group FE	Yes	Yes	Yes	Yes	Yes
Mean of dependent variable	44.94	44.94	44.94	44.94	44.94
Observations	70	70	70	70	70
R^2	0.487	0.488	0.502	0.489	0.496
Panel B: Diesel					
Rival groups within 5 min	−0.002				
	(0.045)				
HHI within 5 min (thousands)		−0.032			
		(0.054)			
Log(1 + rival outlets within 10 min)			−0.210		
			(0.204)		
Driving time to nearest rival (min)				−0.000	
				(0.088)	
Rival outlets within 2 km					0.015
					(0.075)
City of Brno	0.860	0.802	1.084*	0.850	0.811
	(0.599)	(0.597)	(0.636)	(0.638)	(0.704)
Owner-group FE	Yes	Yes	Yes	Yes	Yes
Mean of dependent variable	46.69	46.69	46.69	46.69	46.69
Observations	46	46	46	46	46
R^2	0.510	0.512	0.519	0.510	0.510

OLS; dependent variable: price in CZK per litre. The columns differ in the concentration measure: (1) rival owner groups within 5 minutes; (2) HHI within 5 minutes, equal weights, in thousands; (3) log of one plus the rival outlets within 10 minutes; (4) driving time to the nearest rival outlet; (5) rival outlets within 2 km. Station controls: dummies for the city of Brno, a motorway service station and a price older than seven days, each dropped where it does not vary. The motorway dummy drops out in both panels, because no price comes from a motorway service station, and the stale-price dummy drops out in Panel B. Panel A includes the stale-price dummy, which identifies a single station; its coefficient is not shown, because a dummy for one station has no meaningful robust standard error. Heteroskedasticity-robust (HC1) standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: ceskybenzin.cz (user reports), price lists of Tank ONO and Globus; OpenStreetMap contributors; own calculations.

Table 8: Fuel prices and local concentration: robustness

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Natural 95						
Rival groups within 5 min	−0.004	−0.028				−0.004
	(0.033)	(0.035)				(0.034)
Rival groups, 5 min, excl. unnamed			−0.012			
			(0.041)			
Rival groups within 10 min				−0.012		
				(0.015)		
HHI, 5 min, predicted demand (thousands)					0.003	
					(0.030)	
Station controls	Yes	Yes	Yes	Yes	Yes	Yes
Owner-group FE	Yes	Yes	Yes	Yes	Yes	Yes
Excl. brand conflicts, outliers	No	Yes	No	No	No	No
SE clustered by municipality	No	No	No	No	No	Yes
Clusters						49
Mean of dependent variable	44.94	44.92	44.94	44.94	44.94	44.94
Observations	70	60	70	70	70	70
R^2	0.487	0.549	0.488	0.498	0.487	0.487
Panel B: Diesel						
Rival groups within 5 min	−0.002	0.029				−0.002
	(0.045)	(0.050)				(0.043)
Rival groups, 5 min, excl. unnamed			−0.017			
			(0.057)			
Rival groups within 10 min				−0.025		
				(0.017)		
HHI, 5 min, predicted demand (thousands)					−0.011	
					(0.058)	
Station controls	Yes	Yes	Yes	Yes	Yes	Yes
Owner-group FE	Yes	Yes	Yes	Yes	Yes	Yes
Excl. brand conflicts, outliers	No	Yes	No	No	No	No
SE clustered by municipality	No	No	No	No	No	Yes
Clusters						38
Mean of dependent variable	46.69	46.39	46.69	46.69	46.69	46.69
Observations	46	38	46	46	46	46
R^2	0.510	0.133	0.510	0.530	0.510	0.510

Only the concentration coefficients are shown; station controls and owner-group fixed effects as in Table 7. (1) Baseline, Table 7, column (1). (2) Without the stations whose brand differs between OSM and the price site and without price outliers (more than four robust standard deviations from the median). In Panel B the R^2 falls, because the three flagged MOL diesel prices in Brno (49.9–51.9 CZK) drive the fit of the fixed effects in the full sample. (3) Rival groups without the unnamed stations from the register import. (4) Rival groups within 10 minutes. (5) HHI within 5 minutes with predicted-demand weights, in thousands. (6) Standard errors clustered by municipality, with the districts of Brno as separate clusters. Robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: as Table 7.

8 Limitations and extensions

- OSM is complete for chain stores but patchy for small independents; 111 petrol stations have no name or brand, and duplicate OSM objects of one station can only be removed by rules.
- Without sales data, shares rest on assumptions; we report three weighting schemes to show how much they matter.
- Isochrones assume free-flowing traffic, so they are too large at rush hour. The road network covers Czechia only, so 40 of the 44 Austrian and Slovak stations count only in the radius definitions.
- The population grid combines the 2021 census with the buildings of 2026, so growth since 2021 within municipalities is spread over all buildings, and the baseline puts much demand into non-residential buildings. There are no customer locations outside the region.
- Crowd-sourced prices cover only part of the stations and can be stale; the price regressions rest on 70 and 46 stations.
- Extensions for clinic projects: an ex-post evaluation of a past Czech merger with local overlaps; a check of the building-based population grid against the official 2021 census grid; daily price data to study price changes after entry.

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A Additional tables

Table 9: Supermarkets: rivals and concentration by catchment definition

Catchment definition	Mean	Median	10th percentile	90th percentile
Rival outlets (outlets of other owners)				
1 km radius	4.9	2	0	18
2 km radius	11.9	6	1	33
3 km radius	20.5	13	1	48
5 min drive	20.7	14	2	48
10 min drive	60.1	69.5	5	102
15 min drive	91.1	96.5	30	127
10 min, hypermarkets 15 min (main)	63.7	71.5	6	107
Rival owner groups				
1 km radius	3.3	2	0	11
2 km radius	6.4	4	1	16
3 km radius	9.4	6	1	21
5 min drive	9.5	6	2	20
10 min drive	20.8	25	5	29
15 min drive	28.9	32	11	34
10 min, hypermarkets 15 min (main)	21.6	26	5	30
HHI, equal weights				
1 km radius	4,722	3,750	1,250	10,000
2 km radius	3,237	2,500	1,263	5,000
3 km radius	2,579	2,000	1,314	5,000
5 min drive	2,486	2,000	1,254	3,889
10 min drive	1,652	1,539	1,382	2,000
15 min drive	1,543	1,542	1,383	1,634
10 min, hypermarkets 15 min (main)	1,641	1,539	1,390	2,000
Driving time to the nearest outlet of another owner (minutes)				
All outlets	1.48	1.15	0.33	3.16

The 152 supermarkets in the Brno metropolitan area. Rival outlets: outlets of other owners in the catchment area; rival owner groups: distinct other owners, every independent counted separately. Percentiles from Stata's summarize, detail. Source: OpenStreetMap contributors (stores 13 October 2026, roads 12 October 2026); own calculations.

Table 10: Petrol stations: rivals and concentration by catchment definition

Catchment definition	Mean	Median	10th percentile	90th percentile
Rival outlets (outlets of other owners)				
1 km radius	1.0	0	0	3
2 km radius	2.6	1	0	7
3 km radius	5.0	3	0	14
5 min drive	6.0	4	0	14
10 min drive	21.7	13	4	59
15 min drive	42.7	27	13	98
Rival owner groups				
1 km radius	0.9	0	0	3
2 km radius	2.3	1	0	6
3 km radius	4.1	3	0	11
5 min drive	4.9	4	0	12
10 min drive	14.1	11	4	33
15 min drive	25.1	20	11	47
HHI, equal weights				
1 km radius	7,101	10,000	2,500	10,000
2 km radius	5,148	5,000	1,429	10,000
3 km radius	3,836	2,500	1,016	10,000
5 min drive	3,113	2,000	1,025	10,000
10 min drive	1,271	1,022	637	2,000
15 min drive	761	679	562	972
Driving time to the nearest outlet of another owner (minutes)				
All outlets	2.26	1.68	0.41	5.53

The 343 petrol stations in the South Moravian Region; competitors include the stations in the 10 km buffer. Definitions as in Table 9. Source: OpenStreetMap contributors (stations 13 October 2026, roads 12 October 2026); own calculations.

Table 11: Price regressions: summary statistics

Variable	Mean	Std. dev.	Min	Max
Natural 95 sample (70 stations)				
Price (CZK per litre)	44.94	0.86	43.50	46.90
Rival groups within 5 min	5.80	4.56	0	18
HHI within 5 min	2,813	2,440	656	10,000
Log(1 + rival outlets within 10 min)	2.94	0.98	0.69	4.34
Driving time to nearest rival (min)	1.91	1.70	0.21	6.96
Rival outlets within 2 km	2.90	2.76	0	12
City of Brno	0.31	0.47	0	1
Price older than 7 days	0.01	0.12	0	1
Diesel sample (46 stations)				
Price (CZK per litre)	46.69	1.47	44.90	51.90
Rival groups within 5 min	5.76	4.31	0	18
HHI within 5 min	2,810	2,561	656	10,000
Log(1 + rival outlets within 10 min)	3.02	0.93	1.10	4.34
Driving time to nearest rival (min)	2.15	1.84	0.18	8.34
Rival outlets within 2 km	2.59	2.50	0	12
City of Brno	0.28	0.46	0	1

Stations in the South Moravian Region with a reported price. HHI with equal weights (0–10,000). Source: ceskybenzin.cz (user reports), price lists of Tank ONO and Globus; OpenStreetMap contributors; own calculations.